Dr. Sudhansu Bala Das

Postdoctoral Researcher
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Summary

Currently at the Department of Data Science, School of Computer Science, **University of Galway**, Ireland, I serve as a Postdoctoral Researcher within the **Insight SFI Research Centre for Data Analytics**, working on the *Global and Local Scholarship on Annotated Manuscripts (GLOSSAM)* project.

My research focuses on applying Artificial Intelligence, Natural Language Processing, Machine Learning, Knowledge Graphs, and Data Science methods to multilingual, low-resource, and historically complex textual data.

My background includes a Ph.D. from **National Institute of Technology Rourkela** focused on Machine Translation for Indian Languages, complemented by experience as an Assistant Professor, government-funded NLP researcher, and Research Fellow at **DRDO**, India.

Current Research Position (February 2025 – Present)

Postdoctoral Researcher

Affiliation: Department of Data Science, School of Computer Science, **University of Galway**, Galway, Ireland

Research Centre: **Insight SFI Research Centre for Data Analytics**

Project: Global and Local Scholarship on Annotated Manuscripts (GLOSSAM)

Principal Investigators / Supervisors: **Prof. Pádraic Moran** and **Prof. Edward Curry**

Role: Data Scientist and AI Researcher

- Developing Artificial Intelligence (AI), Natural Language Processing (NLP), and Data Science methodologies for the GLOSSAM project, focusing on computational analysis, knowledge discovery, and intelligent processing of multilingual medieval manuscript collections.
- Working as a Data Scientist on the MÍRA (Manuscripts with Irish Associations) project, designing semantic data models and transforming heterogeneous manuscript metadata into structured knowledge representations using RDF, ontologies, and Linked Open Data (LOD) technologies.
- Developing scalable NLP and machine learning pipelines for information extraction, entity recognition, semantic modelling, metadata enrichment, and automated knowledge graph construction from complex historical datasets.
- Applying graph data science techniques, including network modelling, centrality analysis, clustering, community detection, and relationship mining to identify hidden patterns across manuscripts, people, places, institutions, and historical collections.
- Building reproducible end-to-end research workflows integrating TEI/XML processing, Python-based data engineering pipelines, RDF/OWL knowledge representation, graph analytics, and interactive data visualisation.

Research Interests

Natural Language Processing; Machine Translation; Large Language Models; Low-resource Languages; Digital Humanities; Knowledge Graphs; Network Analysis; Information Extraction; Explainable AI; Multimodal AI; Data Science; Human–Computer Interaction.

Involvement in Funded Research Projects

Project	Contribution
Global and Local Scholarship on Annotated Manuscripts (GLOSSAM)	**University of Galway, Ireland.** Developing Artificial Intelligence (AI), Natural Language Processing (NLP), machine learning, knowledge graph construction, and network analysis methodologies for computational analysis and knowledge discovery from multilingual medieval manuscript collections.
Manuscripts with Irish Associations (MÍRA)	**University of Galway, Ireland.** Working as a Data Scientist to design semantic data models and transform heterogeneous manuscript metadata into structured knowledge representations using RDF, ontologies, and Linked Open Data (LOD). Applying graph analytics, network science, and visualisation techniques to explore relationships among manuscripts, places, people, institutions, and historical collections.
Ministry of Electronics and Information Technology (MeitY) Research Project	**National Institute of Technology Rourkela, India.** Developed Machine Translation and Natural Language Processing (NLP)-based educational technologies, including English–Odia multilingual text-processing workflows for improving accessibility of digital educational content.
Defence Research and Development Organisation (DRDO) Research Projects	**Defence Research and Development Organisation (DRDO), India.** Developed Artificial Intelligence (AI)-enabled software systems, multimodal data-processing frameworks, intelligent interfaces, analytical tools, and operational decision-support systems.

Work Experience

Research Experience

Duration	Role & Organisation	Responsibilities
Sep 2020 – Apr 2023	**Project Scholar** **MeitY Project**, **NIT Rourkela**	Developed Machine Translation and NLP-based educational technologies, including English–Odia language processing workflows.
Sep 2018 – Sep 2020	**Senior Research Fellow** **DRDO**, India	Developed AI-based multimodal data-processing systems, intelligent interfaces, and analytical software solutions.
Sep 2016 – Sep 2018	**Junior Research Fellow** **DRDO**, India	Designed software applications, database systems, analytical tools, and supported testing, validation, and deployment activities.

Academic / Teaching Experience

Duration	Role & Organisation	Responsibilities
Jul 2024 – Nov 2024	**Assistant Professor (Grade I)** **XIM University**, India	Taught undergraduate Computer Science courses including Information Coding Theory and Computer Organization & Architecture. Supervised student projects in Machine Learning, NLP, and Data Science.
Jul 2021 – Jul 2024	**Teaching Assistant** **NIT Rourkela**, India	Supported Computer Science teaching through laboratory sessions, tutorials, student mentoring, assessment, and academic activities.

Education

Year	Degree	Institute / University	CPI
2020–2024	Ph.D., CSE	National Institute of Technology Rourkela, India	8.48/10
2014–2016	M.Tech, CSE	Veer Surendra Sai University of Technology, India	9.34/10
2010–2014	B.Tech, IT	Biju Patnaik University of Technology, India	8.24/10

Ph.D. Thesis

- *Exploring Efficacy of Machine Translation System for Indian Languages*
  Sudhansu Bala Das
  National Institute of Technology Rourkela, India
  Duration: September 2020 – December 2024 (Submitted: July 18, 2024; Defended: December 14, 2024; Degree Awarded: December 18, 2024)
  Supervisor: Dr. Tapas Kumar Mishra (NIT Rourkela); Co-supervisor: Dr. Bidyut Kumar Patra (IIT BHU, Varanasi).

Journal Publications

- **Statistical Machine Translation for Indic Languages**. **Sudhansu Bala Das**, Divyajyoti Panda, Tapas Kumar Mishra, Bidyut Kr. Patra. *Natural Language Processing*, Cambridge University Press, Vol. 31(2), pp. 328–345, 2025. **SCI; Impact Factor: 2.3**
- **Multilingual Neural Machine Translation for Indic to Indic Languages**. **Sudhansu Bala Das**, Divyajyoti Panda, Tapas Kumar Mishra, Bidyut Kr. Patra, Asif Ekbal. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, Vol. 23(5), Article 65, pp. 1–32, 2024. **SCI; Impact Factor: 2.0**
- **Improving Multilingual Neural Machine Translation System for Indic Languages**. **Sudhansu Bala Das**, Atharv Biradar, Tapas Kumar Mishra, Bidyut Kr. Patra. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, Vol. 22(6), Article 169, pp. 1–24, 2023. **SCI; Impact Factor: 2.0**
- **An Efficient Average Execution Time–Round-Robin Scheduling Algorithm**. **Sudhansu Bala Das**, S.K. Mishra, A.K. Sahu. *International Journal of Information Technology*, Springer, Vol. 14, pp. 863–876, 2022. **Scopus Indexed**
- **Alarm Coloring and Grouping Algorithm for Root Cause Analysis**. **Sudhansu Bala Das**, Sugyan Kumar Mishra, Anup Kumar Sahu. *Journal of King Saud University - Computer and Information Sciences*, Elsevier, Vol. 33(5), pp. 572–579, 2021. **SCI; Impact Factor: 6.9**

Conference Publications

- **Investigating the Effect of Backtranslation for Indic Languages**. **Sudhansu Bala Das**, Samujjal Choudhury, Tapas Kumar Mishra, Bidyut Kr. Patra. *First Workshop on NLP for Indo-Aryan and Dravidian Languages (IndoNLP 2025), COLING*, Abu Dhabi, United Arab Emirates, 2025.
- **A Thresholding Method for Improving Translation Quality for Indic MT Task**. **Sudhansu Bala Das**, Leo Raphael Rodrigues, Tapas Kumar Mishra, Bidyut Kr. Patra. *Workshop on Advancing NLP for Low-Resource Languages (LowResNLP 2025), RANLP*, Varna, Bulgaria, 2025.
- **Proceedings of the Workshop on Beyond English: NLP for All Languages in an Era of Large Language Models**. **Sudhansu Bala Das**, Pruthwik Mishra, Alok Singh, Shamsuddeen Hassan Muhammad, Asif Ekbal, Uday Kumar Das. *GlobalNLP 2025 Workshop, RANLP*, Varna, Bulgaria, 2025.
- **NIT Rourkela Machine Translation System Submission to WAT 2022 for MultiIndicMT**. **Sudhansu Bala Das**, Atharv Biradar, Tapas Kumar Mishra, Bidyut Kr. Patra. *9th Workshop on Asian Translation (WAT 2022), COLING*, Gyeongju, Republic of Korea, 2022. **CORE A***
- **A Graphical User Interface Based Speech Recognition System Using Deep Learning Models**. **Sudhansu Bala Das**, Deepshikha Swain, Dipak Das. *International Conference on Advances in Distributed Computing and Machine Learning (ICADCML)*, India, 2022. **Best Paper Award**

- **Study of Atmospheric Attenuation of IR Signature of Airborne Vehicle**. R.K. Dey, S. Mahakud, **Sudhansu Bala Das**, P. Roy, D. Das. *International Conference on Range Technology (ICORT)*, Chandipur, India, 2019.
- **Intelligent Optical Tracking**. **Sudhansu Bala Das**, Dipak Das, P.K. Swain. *International Conference on Intelligent Computing and Remote Sensing (ICICRS)*, Bhubaneswar, India, 2019.

Patents

- **Apparatus for Real-Time Prisoner Monitoring & Alerting System Using IoT**. **Sudhansu Bala Das**, et al. *Australian Patent*, 2021.

  Developed an IoT-enabled intelligent monitoring system integrating sensor-based data acquisition, real-time tracking, automated alert generation, and intelligent decision-support mechanisms for safety-critical environments.

Book Chapters

- **Sentiment Analysis for Prediction of Bitcoin Response**. **Sudhansu Bala Das**, Dipak Das. *The Role of IoT and Blockchain: Techniques and Applications*, Apple Academic Press, 2022.
- **Proof-of-Driving Incentive Mechanism for VANET to Enhance Traffic Safety and Efficiency**. Surjya Kanta Daimary, Sekh Main Hasan, Hemanta Kumar Kalita, **Sudhansu Bala Das**. *Data-Driven AI: A Multidisciplinary Approach*, CRC Press, 2026.

Technical Skills

AI & NLP	Machine Learning, Deep Learning, NLP, Machine Translation, LLMs, Transformers, Information Extraction, OCR, Explainable AI.
Programming	Python, Java, C/C++, C#, MATLAB, SQL, LaTeX.
Frameworks	PyTorch, TensorFlow, Hugging Face, Scikit-learn, SpaCy, Pandas, NumPy, NetworkX.
Data & Systems	TEI/XML, JSON, CSV, SQL, Knowledge Graphs, Network Analysis, Data Visualisation.

Academic Service

Organising Committee Member: GlobalNLP 2025, RANLP 2025, Varna, Bulgaria.

Reviewer: IEEE, ACM, Springer, Elsevier, PLOS ONE, Neurocomputing, Complex & Intelligent Systems; RANLP, LowResNLP, GlobalNLP, ICONIP, ICORT, COCOLE, ICADCML.

Key Achievements

- **MSCA Seal of Excellence**, Marie Skodowska-Curie Postdoctoral Fellowship, European Commission, 2026.
- **Best Localization Blog Paper**, SAMVAD 2024.
- **AI Excellence Award**, SAMVAD 2023.
- **ACM India Anveshan Setu Fellowship**, 2023.
- **Best Paper Award**, ICACNI 2022, NIT Warangal.
- **Australian Patent:** Real-time Prisoner Monitoring System, 2021.
- **University Gold Medalist**, M.Tech Computer Science, 2016.

Date: June 17, 2026
Place: Ireland

Dr. Sudhansu Bala Das